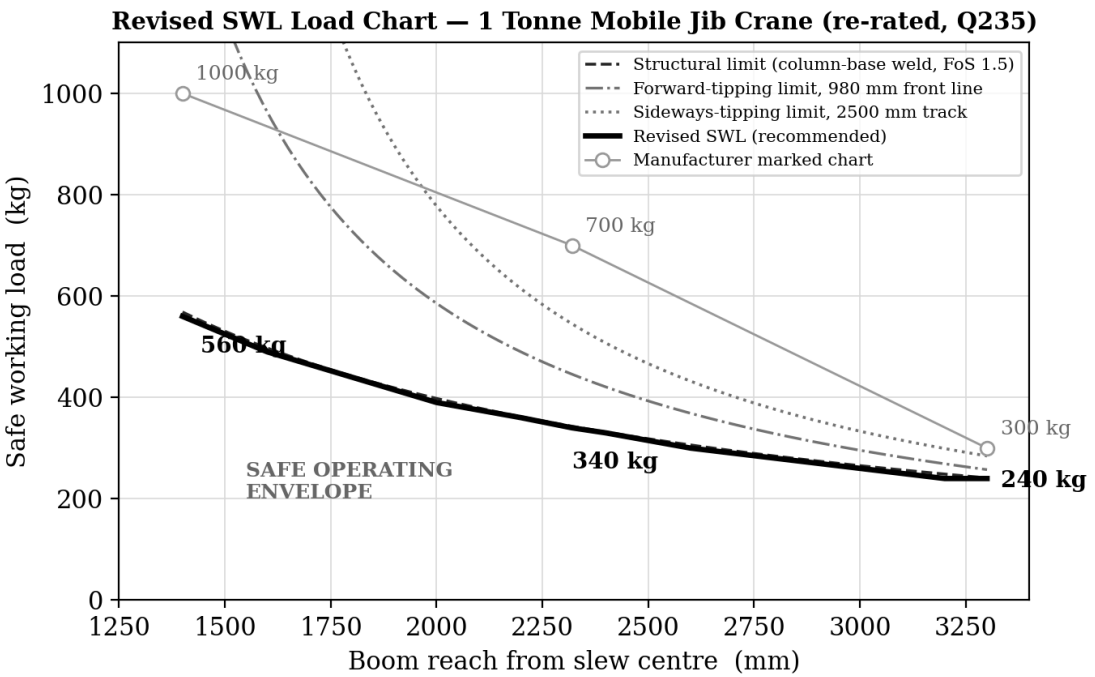


PROJECT 40467361

LOAD RE-RATING REPORT

1 Tonne Mobile Floor Jib Crane

Revised Safe Working Load chart and operating limitations — Q235 material



Revised SWL chart on Q235. The reduced envelope (solid line) restores the required margins across the full operating range; the manufacturer marked chart (open markers) is shown for comparison. The column-base weld governs throughout.

Field	Detail
Project reference	40467361
Subject	Re-rating of the rated load chart to meet working-stress factors of safety on Q235
Material	Q235 carbon structural steel (fy 235 MPa)
Configurations covered	Wheels-only (stowed) and outriggers-deployed (# arrangement)
Original rated capacity	1000 kg at 1400 mm reach (manufacturer marked chart)
Revised maximum SWL	560 kg (governed by the column-to-base weld)
Document revision	Rev R2, supersedes Rev R1 on the Q235 material and 2500 mm track
Date	12 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric verification model, project 40467361)

Document Control

This report converts the findings of the structural verification of the 1 tonne mobile jib crane into a reduced, defensible Safe Working Load (SWL) chart, on the Q235 material. It is a companion to the combined structural verification report, which established that the crane, as marked, does not hold the agreed factors of safety across its rated chart. On Q235 eight of the fourteen structural and mechanical checks fall short on a working-stress basis, clustered on the slewing column and its base weld. This report does not re-derive those checks; it scales their verified results to the target factors of safety and reports the load chart that results.

Revision history

Rev	Date	Description	Status
R1	10 Jun 2026	First issue of the re-rating study on S275, maximum SWL 650 kg.	Superseded
R2	12 Jun 2026	Re-rate recomputed on Q235 (fy 235 MPa, was S275). Weld governs throughout; maximum SWL reduced to 560 kg, with 340 kg at mid and 240 kg at far reach. Deployed track updated to 2500 mm. Proof-load references removed.	Issued for review

Basis and limitations

- **Material.** The crane material is Q235 (GB/T 700), fy 235 MPa, confirmed by the client, in place of the S275 assumed in Rev R1. All sections are 6 mm wall, so the full 235 MPa applies. This lowers the governing column-base weld factor from 0.85 to 0.73 and so deepens the re-rate.
- **Source results.** Factors of safety, member stresses and design forces are taken from the verified parametric model (project 40467361) on Q235. The verification in Section 7 reproduces the published governing-case figures.
- **Re-rating principle.** Every structural demand on this crane scales with the lifted load. The rated load at each reach is reduced until the most heavily loaded check reaches its target factor of safety: 1.5 against yield for members and welds, 3.0 for pins, and 1.5 against overturning.
- **Governing item.** The reduction is set by the column-to-base weld, which carries the lowest working-stress factor (0.73 on Q235) of any load-proportional structural item. On the weaker steel the weld now governs across the whole reach range, including the far end, which the S275 chart had held by forward tipping.
- **Configuration independence.** The governing items sit above the slewing bearing and are fixed by the boom and the lifted load, not by the stance. The revised chart applies in both the wheels-only and the deployed configurations. Deployment remains a precondition for slewing.
- **Not a design certificate.** This is an interim re-rating of an existing unit, not a design. It does not replace a load test or a competent-person examination, which are carried out separately with the approval officer before the crane is returned to work, and it does not remove the need for the structural upgrade if the full 1 tonne rating is to be retained.

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1. Executive Summary

The 1 tonne mobile jib crane, as marked, does not hold the safety margins expected of a rated lifting appliance, and on the Q235 material the shortfall is deeper than the earlier S275 study showed. The independent verification found that eight of fourteen working-stress checks fall short of target on Q235, and that the shortfalls cluster on one load path: the slewing column and the fillet weld at its base. At the governing operating point, 700 kg held out at 2320 mm, the column reaches its yield stress (factor of safety 0.85) and the base weld is past target at 0.73.

This report sets a reduced load chart that restores the required margins. The rated load at each reach has been reduced until the most heavily loaded check, the column-base weld, reaches its 1.5 target. The result is summarised below and given in full in Section 5.

Proposed reduced capacity. The maximum Safe Working Load is reduced to 560 kg (from 1000 kg, and from 650 kg on the earlier S275 chart). At the three marked chart points the revised values are:

Reach (mm)	Marked SWL (kg)	Revised SWL (kg)	Retained	Governed by
1400 (close)	1000	560	56 %	column-base weld
2320 (mid)	700	340	49 %	column-base weld
3300 (far)	300	240	80 %	column-base weld

- One re-rate covers both stances. The governing items sit above the slewing bearing and are fixed by the boom and the lifted load. Deploying the outriggers does not relieve them. The revised chart applies whether the crane stands on its wheels or on its legs. Deployment is still required before any slew.
- Weld governs throughout. On Q235 the column-base weld is the binding limit at every reach, including the far end. On the stronger S275 the far reach had been held by forward tipping; the weaker weld now binds first everywhere, which makes the chart shape simpler but lower.
- No fabrication is required. The re-rate is achieved by limiting the load, not by modifying the crane. It is the honest reflection of what the Q235 structure as built safely carries for repeated duty.
- It is an interim measure. If the full 1 tonne chart must be retained, the column section and its base weld must be upgraded; the re-rate does not remove that need.

At the revised SWL every working-stress check meets its target. Section 7 verifies this: the governing column-base weld sits at 1.50, the column rises to 1.75, and stability, the boom, the outrigger members and the pins all clear with margin. The only item that does not clear on the conservative assumption is the cylinder rod-end pin if it is the 20 mm pin; the 40 mm Detail-4 pin, which the drawings indicate is the fitted part, clears comfortably and must be confirmed.

2. Background and Basis

The unit is a mobile floor jib crane rated to lift 1 tonne. It runs on four wheels, carries a counterweight at the rear, and luffs a three-stage telescopic boom on a single hydraulic cylinder. The upper assembly slews on a bolted bearing flange above the base frame. The manufacturer marks three rated points, with capacity falling as reach grows: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm.

The verification was carried out by two independent methods against the same geometry: closed-form hand calculation to Eurocode 3 and EN 13001, and a small in-script plane-frame finite element model, both on the Q235 material. The governing design moment at the column is 19.91 kN·m, produced not by the heaviest lift but by the 700 kg load at 2320 mm, because the long lever arm drives the structure hardest.

2.1 The checks that fall short

In the wheels-only state, eight of fourteen checks do not meet target on Q235. They are listed below with the load fraction each would need, relative to the governing-case demand, to reach its target. The smallest fraction is the binding constraint.

Check	FoS now	Target	Load fraction to target	Binds?
Weld, column-to-base	0.73	1.50	0.49	GOVERNS
Column, combined N+M	0.85	1.50	0.57	no
Weld, outrigger	1.11	1.50	0.74	no
Outrigger arm	1.30	1.50	0.87	no
Boom outer (100x140x6)	1.26	1.50	0.84	no
Pivot pin Ø30	2.08	3.00	0.69	no
Stability, forward (worst)	0.96	1.50	see 4	no
Cylinder pin (Ø20 as modelled)	1.03	3.00	0.34	flagged*

* The cylinder rod-end pin is carried at the conservative 20 mm diameter. The only hinge pin shown near the cylinder mount is the 40 mm Detail-4 pin, which clears its check at 4.11 on Q235. The as-built diameter must be confirmed; the re-rate assumes the 40 mm pin.

2.2 Why the column-base weld governs

Reducing the load reduces the column-base moment in direct proportion, and the weld stress with it. Because the weld starts from the lowest factor of safety of any load-proportional structural item (0.73 on Q235), it reaches the 1.5 target at a lower load than the column does. Holding the load to where the weld clears 1.5 leaves the column at 1.75 and every other structural member above target. The weld is therefore the single item that sets the chart, and it sets it identically in both stances, since it lies above the slewing bearing.

3. Re-Rating Methodology

The method is a direct, transparent scaling of the verified results. It has three steps.

1. Express every demand as a function of the lifted load and reach. The column-base moment is $M = 1.25 \times m \times g \times r$, where m is the lifted mass, g is gravity, r is the reach and 1.25 is the HC1 dynamic factor. The column, the base weld, the boom and the cylinder pins all scale with this moment. The wheel reaction and the outrigger members scale with the load and the geometry. Stability is a moment balance on the static load.

2. Find the binding constraint and its limiting moment. On Q235 the column-base weld reaches its 1.5 target when its stress falls to 115.1 MPa, which corresponds to a column-base moment of 9.76 kN·m, against the 19.91 kN·m produced by the marked chart at mid reach.

3. Convert the limiting moment to an allowable load at every reach, then verify the rest. The allowable mass at reach r follows from $m = M_{\text{allow}} / (1.25 \times g \times r)$. Forward stability, sideways stability and the pins are then checked at those loads. On Q235 the weld is the binding item at every reach, so the chart follows the weld limit throughout and no stability trim is needed.

Governing calculation (column-to-base weld, target FoS 1.5, Q235)

```
weld stress now sigma_w = 234.9 MPa at M = 19.91 kN.m (FoS 0.73 on Q235)
weld FoS at marked load = 0.7 x fu / (1.5 x sigma_w) basis -> 0.73
limiting moment M_allow = 19.91 x (0.73 / 1.50) = 9.76 kN.m
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Allowable lifted mass at reach r :

$$m_{\text{allow}}(r) = M_{\text{allow}} / (1.25 \times g \times r) = 9\,760 / (12.2625 \times r) \quad [\text{kg}, r \text{ in m}]$$

$$r = 1.40 \text{ m} \rightarrow 569 \text{ kg} \quad r = 2.32 \text{ m} \rightarrow 343 \text{ kg}$$

$$r = 2.80 \text{ m} \rightarrow 284 \text{ kg} \quad r = 3.30 \text{ m} \rightarrow 241 \text{ kg}$$

The full set of equations is evaluated numerically across the reach range to produce the chart in Section 5. The numbers above are the closed-form check on the binding item.

4. Governing Constraints Across the Range

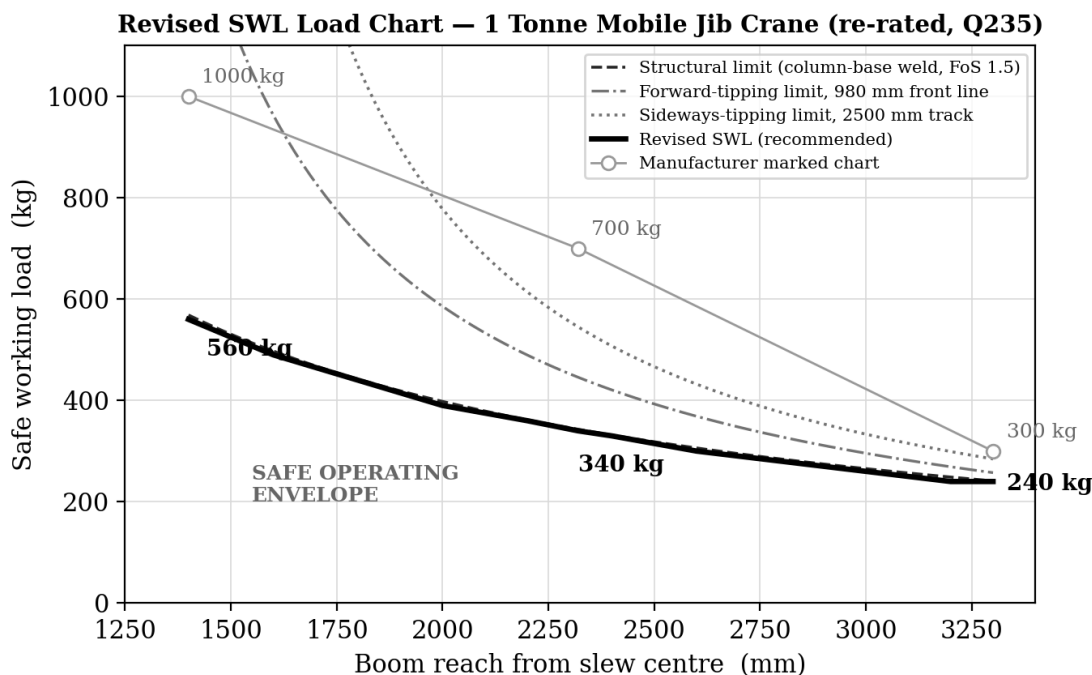
Three families of limit act on the crane. On Q235 their interaction is simpler than on S275: the weld now governs everywhere.

Limit	Behaviour and role in the re-rate
Structural (column-base weld)	Scales with the column-base moment. On Q235 it binds across the whole range, including the far end, and is the same in both stances. It sets the entire revised chart.
Forward stability	Moment balance about the front foot line, 980 mm ahead of the column, in both stances. At the reduced loads it clears 1.5 everywhere (worst about 1.97 at mid reach), so it does not bind.
Sideways stability (slewing)	Applies only when the boom is slewed across the rails on the deployed legs. On the 2500 mm track it clears 1.5 at the reduced loads (worst about 2.41 at mid reach), so it does not bind.
Pins (Ø40 confirmed)	Scale with the moment. With the 40 mm Detail-4 cylinder pin and the 30 mm pivot pin, all pins clear the 3.0 target at the reduced loads with wide margin. The 20 mm cylinder pin would bind; see Section 9.

Because the weaker Q235 weld governs the close, mid and far reaches alike, and is configuration-independent, a single revised chart serves both stances. Unlike the S275 chart, where the corrected 980 mm front line made forward tipping co-govern the far reach, on Q235 the weld binds first even there, so no far-reach stability trim is applied.

5. Revised SWL Load Chart

The chart below gives the maximum allowable load at each reach across the full operating range. The continuous curve is the computed weld limit; the recommended SWL is rounded down to the nearest 10 kg for use on the machine.



Revised SWL chart on Q235. The solid black curve is the recommended SWL, with the safe operating envelope shaded. The dashed line is the structural (column-base weld) limit, which governs throughout; the dotted and dash-dot lines are the sideways- and forward-tipping limits, both of which sit above the weld limit on Q235. The thin line with open markers is the manufacturer marked chart.

5.1 Maximum allowable load at each reach

Intermediate reaches may be read from the chart or interpolated linearly between the tabulated points (always rounding down).

Reach (mm)	Computed limit (kg)	Recommended SWL (kg)	Governing limit	vs marked
≤ 1400	569	560	weld / max SWL cap	56 %
1600	497	490	column-base weld	—
1800	442	440	column-base weld	—
2000	398	390	column-base weld	—
2200	362	360	column-base weld	—
2320 (mid)	343	340	column-base weld	49 %
2400	332	330	column-base weld	—
2600	306	300	column-base weld	—
2800	284	280	column-base weld	—

Reach (mm)	Computed limit (kg)	Recommended SWL (kg)	Governing limit	vs marked
3000	265	260	column-base weld	—
3200	249	240	column-base weld	—
3300 (far)	241	240	column-base weld	80 %

Reading the chart. The maximum SWL is 560 kg and applies at any reach up to 1400 mm. Beyond 1400 mm the allowable load falls along the weld limit. On Q235 the column-base weld governs at every reach, in both stances, so the chart is a single curve with no stability trim. The chart never exceeds 560 kg.

6. Proposed Reduced Capacity (summary)

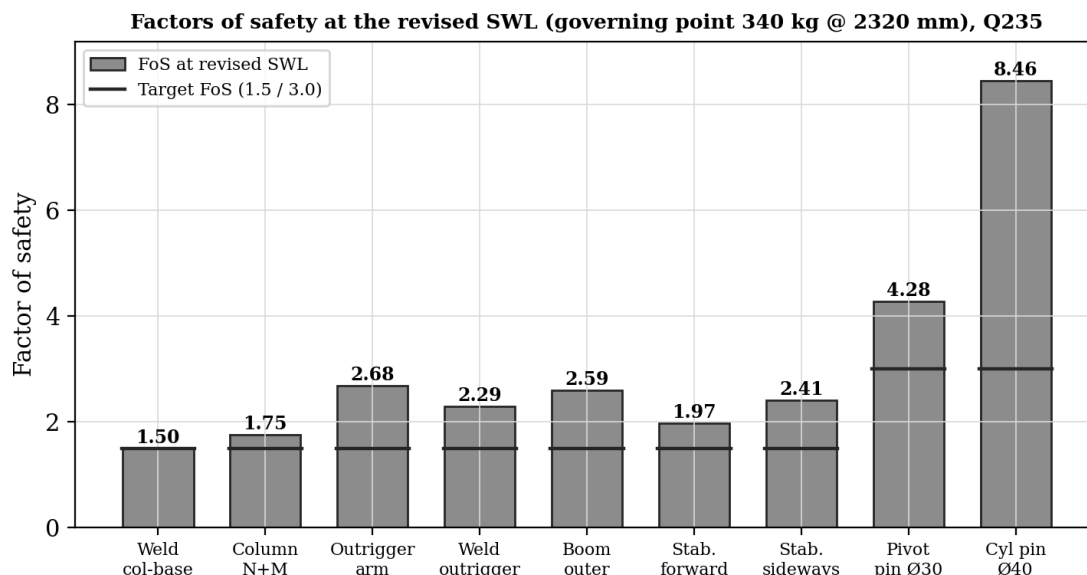
For the duty plate and the operating manual, the revised rating is stated plainly below.

Item	Revised value
Maximum Safe Working Load	560 kg (at reach $\leq$ 1400 mm)
SWL at 1400 mm reach	560 kg
SWL at 2320 mm reach	340 kg
SWL at 3300 mm reach	240 kg
Material	Q235 (fy 235 MPa)
Basis	Working-stress, FoS 1.5 members and welds, 3.0 pins, 1.5 overturning
Dynamic factor included	1.25 (HC1 hoist duty)
Applies in	Both wheels-only and deployed configurations
Slewing	Permitted only with outriggers deployed; see Sections 8 and 9

A revised duty plate carrying the three reduced chart points and the 560 kg maximum should be fitted to the machine, and the existing 1 tonne chart should be obscured or removed, before the crane is returned to service under this re-rate.

7. Verification at the Revised SWL

The figure and table below confirm that, at the recommended loads, every working-stress check meets its target on Q235. The model was first validated against the published governing-case results; it reproduces them exactly (column 0.85, base weld 0.73, boom outer 1.26, outrigger arm 1.30, stability 0.96, pins 2.08 / 1.03 / 4.11), which gives confidence in the re-rated figures.



Factors of safety at the governing revised point (340 kg at 2320 mm), Q235. Every bar meets or exceeds its target (black tick). The cylinder pin is shown at the 40 mm Detail-4 diameter.

7.1 Factors of safety at the revised SWL

Values below are at the governing mid-reach point (340 kg at 2320 mm). The minimum across the whole revised chart is given where it is lower.

Check	FoS now*	FoS revised	Min on chart	Verdict
Weld, column-to-base	0.73	1.50	1.50	Pass
Column, combined N+M	0.85	1.75	1.74	Pass
Boom outer (100x140x6)	1.26	2.59	2.55	Pass
Outrigger arm	1.30	2.68	2.30	Pass
Weld, outrigger	1.11	2.29	1.95	Pass
Stability, forward	0.96	1.97	1.74	Pass
Stability, sideways (2500 track)	1.17	2.41	2.00	Pass
Pivot pin Ø30	2.08	4.28	4.00	Pass
Cylinder pin Ø40 (Detail-4)	4.11	8.46	8.30	Pass
Cylinder pin Ø20 (conservative)	1.03	2.12	2.08	Confirm

* FoS now is at the marked governing load (700 kg at 2320 mm). The single non-pass is the conservative 20 mm cylinder pin, a confirm-the-diameter item, not a structural shortfall of the re-rate: the 40 mm Detail-4 pin clears at 8.30.

8. Wheels-Only and Deployed Configurations

The revised chart is a single chart that holds in both stances, but the two stances are not interchangeable in service. The distinction is about slewing, not about lifting capacity.

- Lifting capacity is the same in both stances. The column, the base weld and the boom are loaded by the boom and the lifted load, above the slewing bearing, and do not change when the base stands on legs instead of wheels. The reduced chart applies as drawn in both cases.
- Forward (luffing-plane) stability is the same in both stances. The legs are transverse, so they do not move the front tipping line. At the reduced loads forward tipping clears 1.5 in both stances.
- Slewing requires the legs down. On the wheels the track is too narrow to slew safely. Deploying the legs widens the lateral track to 2500 mm and makes slewing possible. The warning plate already requires this, and it remains a hard rule under the re-rate.
- Sideways stability when slewing depends on the deployed track. On the 2500 mm track it clears 1.5 at the reduced loads, so it does not trim the chart. Setting the track wider still adds margin but is not required by the re-rate.

In short: the revised chart governs how much may be lifted; the configuration rules govern whether the crane may slew. Both must be observed.

9. Operating Limitations and Restrictions

The following limitations apply to the crane operated under this re-rate. They should be reflected on the duty plate and in the operating procedure.

9.1 Load and reach

- Maximum SWL 560 kg, at any reach up to 1400 mm. Never exceed the revised chart at any reach.
- Beyond 1400 mm, the load must be reduced along the revised chart of Section 5. Where reach is read between tabulated points, interpolate and round down.
- The reduced rating includes the 1.25 HC1 dynamic factor. Lifts must remain slow and controlled; the factor does not cover snatch loading.

9.2 Slewing and outriggers

- Deploy all four outrigger legs before slewing. Never slew on the wheels.
- Confirm the deployed lateral track on the machine. The drawing shows 2500 mm, which clears sideways stability at the reduced loads. Confirm the achievable deployed track.
- Confirm the leg deployment angle. The product photograph appears to show the front legs splayed forward; if so, forward stability is better than reported. The square (90 degree) assumption used here is the conservative case.

9.3 Connection to confirm

- Confirm the cylinder rod-end pin diameter. The re-rate assumes the 40 mm Detail-4 hinge pin, which clears the 3.0 pin target at 8.30 at the revised loads. If the fitted pin is found to be 20 mm, it must be upsized to 40 mm, or the chart must be reduced further until it is.

9.4 Serviceability and scope

- Long-reach positioning is limited by deflection. Boom-tip deflection at full reach exceeds the L/200 serviceability guide; this is a positioning-accuracy limitation, not a strength one.
- Static lifting only. No wind, side pull, seismic or snatch loading; these are outside the verified scope.
- Interim status. The re-rate does not replace a load test or a competent-person examination, which are carried out separately with the approval officer, and does not authorise the full 1 tonne chart.

10. Conclusions and Path to Full Rating

On a working-stress basis the crane as marked cannot be certified to its 1 tonne chart in either stance, because on Q235 the column reaches yield and its base weld is well past target at mid reach. Reducing the rated load restores the margins without any fabrication. The revised chart, with a maximum SWL of 560 kg and reduced values at reach, brings every working-stress check to target and is the recommended immediate action.

The re-rate is the cheaper of the two routes and is the honest reflection of what the Q235 steel safely carries for daily duty. It does not preclude the alternative.

Route	Outcome
Adopt the revised chart (recommended)	Immediate, no fabrication. The crane returns to safe service at the reduced, verified rating in both stances. This report is the basis.
Upgrade the column and base weld	Retains the full 1 tonne chart. Requires a larger column section, with the base weld grown to suit and the cylinder pin confirmed or upsized. The only route that preserves capacity, at the cost of fabrication. On Q235 the upgrade is larger than the S275 case would have needed.
Confirm mill yield, cylinder geometry and deployed track	A parallel action. Confirming the actual Q235 mill yield (commonly a little above nominal) and the cylinder anchor geometry would lift the column margin and could let the reduced chart be re-issued at slightly higher values.

Recommendation. Adopt the revised chart as the immediate measure so the crane can be used safely now, fit a revised duty plate, and hold the column upgrade as the path to restoring the full rating. Deploy the legs before any slew, and confirm the cylinder rod-end pin diameter. The crane should not be worked to the original 1 tonne chart.

11. Assumptions and Items to Confirm

The re-rate inherits the inputs of the parent reports. The following should be confirmed against the built machine before the revised chart is relied on for certification.

- Material, Q235. Client-confirmed, fy 235 MPa nominal. Mill certificates would confirm the actual yield, which commonly runs a little above nominal and would allow a modest increase in the revised chart.
- Cylinder rod-end pin diameter. Assumed 40 mm (Detail-4). If 20 mm, see Section 9.3.
- Crane self-weight, 700 kg, and CG offset, 0.30 m behind the column. Client-supplied and estimated respectively. A swing of plus or minus 100 kg moves the stability margins; the structural (weld) limit is unaffected.
- Deployed lateral track, 2500 mm. Read off the updated general-arrangement drawing. The as-built deployed width should be measured. On Q235 the weld governs even at far reach, so the track does not set the chart, but it must still be wide enough to clear sideways stability.
- Weld throat, 6 mm. Taken as the drawing minimum. If site welds are smaller the governing weld worsens and the chart must be reduced.
- Front overhang to the tipping line, 980 mm. Used for stability. The warning-plate 420 mm dimension should be reconciled against this on the machine.

12. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- GB/T 700, carbon structural steels (Q235)
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion files

- Combined structural verification report (Rev 5, Q235), the parent document for all carried-over results.
- MATLAB R2026a parametric model (project 40467361), on Q235, which produced the verified factors of safety scaled in this report.
- Console log of the verification run, including the factor-of-safety tables and the column plastic-capacity check.